

Addressing Systematic Non-response Bias with Supervised Fine-Tuning of Large Language Models: A Case Study on German Voting Behaviour

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Abstract

A major challenge for survey researchers is dealing with missing data, which restricts the scope of analysis and the reliability of inferences that can be drawn. Recently, researchers have started investigating the potential of Large Language Models (LLMs) to role-play a pre-defined set of “characters” and simulate their survey responses with little or no additional training data and costs. Previous research has mostly focused on zero-shot LLM predictions. However, often other survey responses are at least partially available. This work investigates the viability and robustness of supervised fine-tuning on these responses to simulate systematic and random item-level non-responses in the context of German voting behaviour.

Our results show when systematic item non-responses are present, fine-tuned LLMs outperform traditional classification approaches on survey data. Fine-tuned LLMs also seem to be more robust to changes in the set of features that the model can use to make predictions. Finally, we see that fine-tuned LLMs match the performance of traditional classification methods when survey responses are missing completely at random.

1 Introduction

For decades the empirical social sciences have relied on survey samples of individuals taken from well-defined populations as one of their main data sources. However, survey-based research is witnessing transformative changes. Among the challenges faced are a declining willingness of people to participate in surveys and the peculiarities of new modes of survey participation, e.g., via the web or smartphones (Berinsky, 2017; Couper, 2017). For surveys about political opinions and voting behaviour, prior research has shown that these biases can affect election forecasts. For instance, in the United States, Republicans are less likely to

respond to follow-up surveys and are less cooperative with phone interviews compared to Democrats (Feldman, 2022; Clinton et al., 2022).

In this context, Large Language Models (LLMs) present a promising alternative, since their ability to generate text can make them operate as human-like dialogue agents that can be imbued with a “personality”, retain memory of facts or interactions, and adaptively respond to social stimuli (Acerbi and Stubbersfield, 2023; Park et al., 2023; Shanahan et al., 2023; Wang et al., 2023). However, LLMs face specific challenges when used to predict the vote choices of selected populations, particularly in zero-shot settings. Recent studies have revealed political biases in LLMs, often leaning left on political orientation tests (Santurkar et al., 2023; von der Heyde et al., 2024; Bisbee et al., 2024). Additionally, zero-shot LLMs face technical challenges such as answer extraction problems, where first-token probabilities do not match the actual text responses (Wang et al., 2024), and prompt stability issues, where small changes in prompt wording can lead to inconsistent responses (Bisbee et al., 2024). The effectiveness of in-context survey prediction with LLMs, in general, has been questioned, as this approach struggles to replicate the variation and statistical relationships found in human survey data (Dominguez-Olmedo et al., 2023). This suggests that zero-shot approaches may not be sufficient for predicting survey responses.

Advances in the computational efficiency of fine-tuning, due to model quantization and low-rank adaptation (Dettmers et al., 2023), have made fine-tuning a more accessible alternative, with the potential to mitigate these issues (Liu et al., 2022). Furthermore, the main problem with fine-tuning – the limited ability to fine-tune state-of-the-art proprietary models – is offset by recent performance improvements of open-source models (Dubey et al., 2024), and the added benefit of being able to deploy the model locally, addressing challenges with

confidential survey data and reproducibility.

Kim and Lee (2023) showed that fine-tuning LLMs can be effective for imputing survey data in the American context. In this work, we focus on the German context instead. We impute¹ the “vote choice” variable from the German Longitudinal Election Study (GLES) post-election cross-section (GLES, 2020). Previous research on this dataset has shown that zero-shot LLMs are able to capture broad tendencies tied to partisanship, but missed out on more nuanced factors that impact individual voter choices (von der Heyde et al., 2024).

To address this limitation we fine-tune a range of open-source LLMs with different subsets of the GLES and evaluate their performance on the “vote choice” for held-out participants. We compare the LLMs’ performance against various baseline methods. We do not only focus on predicting responses that were missing at random but also on systematic non-responses. Systematic non-responses lead to non-response bias, meaning the survey may not accurately reflect the target group due to differences between respondents and non-respondents. Our work addresses the following research questions:

RQ1: Are fine-tuned LLMs more effective and robust in imputing **random survey responses** than established methods and zero-shot LLMs?

RQ2: Are fine-tuned LLMs more effective and robust in imputing survey responses than established methods and zero-shot LLMs when **systematic non-response bias** is present?

We find that (1) fine-tuning LLMs improves their accuracy for vote choice predictions, (2) fine-tuned LLMs produce less biased vote share distributions than zero-shot LLMs, and (3) fine-tuned LLMs exceed the performance of established classification methods in case of systematic non-responses. These results are robust across different models sizes (0.5 to 8B parameters) and model families (Llama-3 and Qwen-2.5). Our results suggest that fine-tuned LLMs are less impacted by the removal of features correlated with the target variable compared to zero-shot LLMs and traditional classification methods.

¹We refer to the process of replacing missing data with substituted or predicted values as imputation. In our work we impute values that closely approximate missing values and frame this problem as a classification task.

2 Related Work

2.1 Opinion Prediction with LLMs

Most prior research on opinion prediction with LLMs has focused on in-context approaches, primarily using zero-shot methods (Argyle et al., 2023; von der Heyde et al., 2024; Dominguez-Olmedo et al., 2023; Bisbee et al., 2024; Santurkar et al., 2023; Qu and Wang, 2024). Studying zero-shot setups is particularly valuable for identifying potential biases LLMs may have acquired during pre-training. There is the potential of fine-tuning on online resources for opinion prediction, as well (Jiang et al., 2022; Chu et al., 2023). However, these studies differ from our work since they (i) use other data sources and mainly focus on the US context, (ii) do not use other participants’ responses to inform the prediction or (iii) use different models.

Data Sources Studies have utilized various datasets to evaluate LLMs in opinion prediction tasks. The American National Election Study (ANES) data was utilized in several studies (Argyle et al., 2023; Bisbee et al., 2024; Jiang et al., 2022). Pew Research Center survey data was employed by Santurkar et al. (2023) and Chu et al. (2023). Other datasets included the American Trends Panel (Santurkar et al., 2023), American Community Survey (ACS) (Dominguez-Olmedo et al., 2023), the World Values Survey (Qu and Wang, 2024), and German Longitudinal Election Study (GLES) data (von der Heyde et al., 2024). For studies involving fine-tuning, Jiang et al. (2022) used Twitter data and Chu et al. (2023) trained on content from various US media outlets.

Models The majority of existing studies relied on closed-source OpenAI models for their experiments, predominantly GPT-3 or GPT-3.5 (Argyle et al., 2023; Santurkar et al., 2023; von der Heyde et al., 2024; Qu and Wang, 2024), with Bisbee et al. (2024) being a more recent study that used GPT-4. Dominguez-Olmedo et al. (2023) evaluate a range of open-source models in-context and all studies involving fine-tuning employed open-source models (Jiang et al., 2022; Chu et al., 2023). The prevalent use of OpenAI models was mainly due to their significantly superior performance compared to open-source counterparts at the time, coupled with their widespread adoption through the popularity of ChatGPT. Recently, however, open-source models have started to close this performance gap (Dubey et al., 2024).

Findings While some approaches using additional context for fine-tuning and partisan pre-training show promise (Jiang et al., 2022; Chu et al., 2023), previous research has found opinion prediction with LLMs to not be reliably enough replace traditional surveys (Bisbee et al., 2024). Generally, previous work has found that LLMs can recover broad average opinions but struggle with nuanced subgroup differences (Argyle et al., 2023; Dominguez-Olmedo et al., 2023; Santurkar et al., 2023). Additionally, this performance degrades for non-US contexts and non-English languages (von der Heyde et al., 2024; Qu and Wang, 2024).

2.2 Imputation with LLMs

Recently, LLMs have been applied in context of multiple data wrangling tasks including imputation (Mei et al., 2021), with the well-known benchmark Holistic Evaluation of Language Models (HELM) (Liang et al., 2023) having a dedicated scenario for imputation (Narayan et al., 2022). Narayan et al. (2022) found that established approaches either cannot or struggle to impute values not seen in the training set. Additionally, they show that fine-tuning is effective for this task. They fine-tune models with 1.3 billion and 7 billion parameters and find that both outperform GPT-3-175B in the zero-shot scenario and score similarly in the few-shot scenario.

Missing data (non-response) problems frequently occur in surveys. For survey imputation specifically, Kim and Lee (2023) found that fine-tuning LLMs on the General Social Survey is more effective than Matrix Factorization for different missing data scenarios, including for data missing not at random. Additionally, they found that the decoder model Alpaca-7b performed better than the encoder model RoBERTa-large.

Ji et al. (2024) use a Retrieval-Augmented Generation (RAG) approach on the American Trends Panel. They find that their approach matched and, in some cases, outperformed established imputation methods when using OpenAI models, whereas smaller open-source models (e.g., Llama-3-8B) performed less effectively.

3 Methods and Experiments

We aim to predict the target variable “vote choice” using survey variables that in the literature have been found to be important predictors of voting behaviour as features. We compare the perfor-

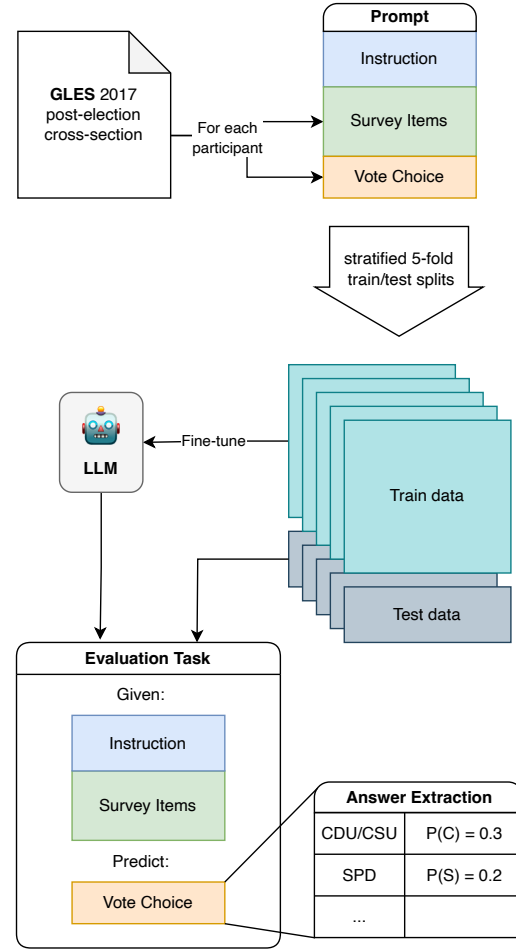


Figure 1: Overview of the method we use to fine-tune and extract answers from the LLM. The prompt design is depicted in more detail in Figure 2. The different training data splits we use for each experiment are depicted in Figure 3.

mance of fine-tuned LLMs to zero-shot LLMs and established classification methods. Figure 1 shows the basic setup of our experiments. We use stratified 5-fold cross-validation for all experiments and compare the performance of different classification models on a randomly selected set of survey participants (test data). To measure performance we use macro-F1 scores, as our models should perform equally well on all parties.² Additionally, we analyse the predicted vote share distributions (normalized sum of output probabilities) of different models and compare it to the vote share distribution of the survey.

We report the cross-validation performance by giving the mean of a metric and the 95% confidence intervals.

²Accuracy and balanced accuracy scores can be found in the appendix.

3.1 Survey Data Processing

We used the German Longitudinal Election Study (GLES) Cross-Section Cumulation 2009-2017 dataset for our experiments (GLES, 2020), which is openly available for academic research. This dataset contains pre- and post-election surveys conducted in face-to-face interviews eight weeks before and after each federal election. In our experiments, we predicted the vote choices of the 2017 post-election cross-section to enable comparison with previous zero-shot results reported by von der Heyde et al. (2024).

Item Selection Survey items were selected based on factors commonly associated with vote choice in German elections (Schmitt-Beck et al., 2022; Schoen et al., 2017; Klein, 2014). These included demographic information (“Age”, “Gender”, “Education”, “Income”, “Employment Status”), “Religiosity”, “Left-Right-Ideology”, “Party ID” (party identification), “Party ID Strength”, “Residency”, and attitudes towards “Immigration” and “Income Inequality”. See Appendix Table 1 for the corresponding survey questions. We use the participant’s “Vote choice” as the target variable for all experiments.

We use the German survey questions and answers instead of translating them. LLMs tend to perform well in German, so a translation is not necessary. Additionally, some cultural undertones might be lost when translated.

Most survey items were mapped one-to-one to the processed data. That is, the chosen survey questions correspond to one feature each, and each survey answer option is taken as a feature value. To reduce the number of classes of the classification task, the infrequently occurring answer choices *invalid vote* and *don’t know* were dropped.

Missing Values Like von der Heyde et al. (2024), we dropped missing values for vote choices. Other missing values in the survey data were retained rather than dropped or imputed. This decision was made because the type of missingness provides relevant information that the LLM might be able to use for better vote predictions. There are a variety of missing values, e.g., *no answer* for when the participant does not answer a question, or *interview aborted* when an interview was aborted. Further, items like *income* have higher rates of missingness, which may indicate participants’ reluctance to share this information.

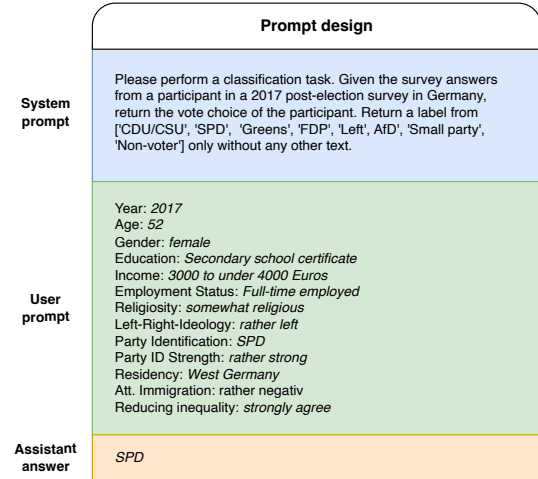


Figure 2: Prompt design. The information in italics is filled with the answers of the respective survey participant. This is an English translation. See Appendix Figure 7 for the original.

3.2 LLM Setup

Prompt Design As we are performing an instruction-based task, we use instruction-tuned models (also called chat models).³ These instructions are split into a system prompt defining the task and a user prompt giving relevant information to an instance of this task. This is followed by the assistant answer, which contains the “vote choice” variable the model should predict.

Figure 2 shows an example of such a prompt. Each prompt includes an instruction specifying our classification task, followed by the survey responses for a specific participant. The “Assistant answer” is the correct vote choice for the given participant.

Model Selection To check robustness of our fine-tuning results we run our experiments on a range of model sizes and two different model families. We use the newest version of the Llama (Dubey et al., 2024) and Qwen (Qwen et al., 2025) families⁴. Both of these model families perform well on benchmarks and offer a wide range of model size to choose from, as well as, first-party instruction-tuning. We select all models that have under 8 billion parameters from those families.

³For brevity, we omit the “Instruct” suffix in model descriptions for the following sections.

⁴Specifically Llama-3.2-1B-Instruct, Llama-3.2-3B-Instruct, Llama-3.1-8B-Instruct, Qwen2.5-0.5B-Instruct, Qwen2.5-1.5B-Instruct, Qwen2.5-3B-Instruct, and Qwen2.5-7B-Instruct.

Supervised Fine-Tuning We train on a 40 GB MIG of an A100 GPU. We use half-precision, i.e., 16 bit weights. To reduce the memory requirement and increase the efficiency of the fine-tuning process we use low-rank adapters (Liu et al., 2022). We set the rank for LoRA to the highest value without running out of memory ($rank = 256$) during the fine-tuning process. We use LoRA on all linear layers with rank stabilisation with $\alpha = 8$ (Kala-jdziewski, 2023). We use a batch size of 1 for memory efficiency. No gradient accumulation was used, as it did not show performance benefits in preliminary testing on an independent dataset. We run each fine-tuning for 3 epochs, all other hyperparameters are set to the default using the Huggingface supervised fine-tuning trainer with training on completions only.⁵

Answer Extraction We extracted the model’s response by selecting the highest probability token at the index of the first vote token, similar to the approach of Argyle et al. (2023). This is a viable extraction option since all vote choice options: CDU/CSU, SPD, Greens, FDP, Left, AfD, small party, and non-voter, start with different first tokens.

Answer extraction is an ongoing field of research and using first token probabilities has weaknesses in zero-shot settings (Dominguez-Olmedo et al., 2023; Wang et al., 2024). For example, a model might be quite likely to answer with text like *Not solvable*, which starts with the same token as *Non-voter* but does not have similar meanings. There is currently no obvious best way to solve this, with multiple options being viable (Fourrier et al., 2023).⁶ This is, however, not a problem after our supervised fine-tuning, as any token log-likelihood that is not in the vote choice options will be insignificantly low after a sufficient amount of training steps.

3.3 Experiments

The goal of our experiments is to compare the performance of different models for predicting the “vote choice” of a randomly selected set of survey participants (test data). Figure 3 shows that we use the same test set for all experiments. We test the

⁵See https://huggingface.co/docs/trl/v0.13.0/en/sft_trainer

⁶See <https://github.com/huggingface/lighteval/tree/7295c78fcb63604cc18c43dd62349a6120ac8600> where our implementation is most similar to loglikelihood_f1_single_token.

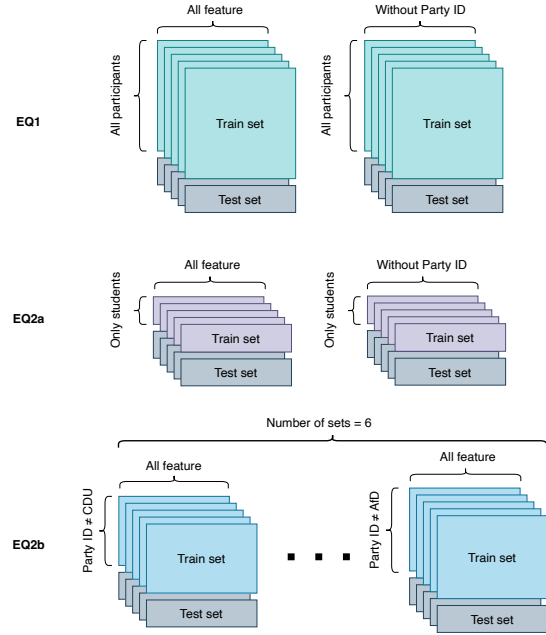


Figure 3: The survey data splits used for each experiment.

robustness of our results by repeating the experiments, when applicable, without the variable “party identification” since party identification is known to be the most important predictor for vote choice (see Section A.3).

RQ1 Experiment In context of the first research question, we investigate if the fine-tuned models outperform zero-shot models and traditional classification methods for a scenario with missing data completely at random (MCAR). We therefore design the following experiment:

EQ1: We split the GLES 2017 post-election cross-section into stratified 5-fold train/test splits. We fine-tune our selected models on the train set with the configuration and validation procedure detailed earlier.

RQ2 Experiments For the second research question, we explore the performance of fine-tuned models when systematic non-response bias is present. We focus on data missing not at random (MNAR) as this type of missingness presents the biggest challenge for traditional imputation methods (Van Buuren, 2018). We design the following scenarios:

EQ2a: We predict systematic non-responses by only training on students (i.e., people enrolled in school or university) leaving out any other kind of occupation from the training set

used in EQ1. This experiment is inspired by the fact that a lot of survey experiments rely on so-called “convenience samples”, i.e., groups that are easy to reach (Mullinix et al., 2015).

EQ2b: We predict systematic non-responses by excluding survey respondents with a certain party identification from the training set used in EQ1. This experiment is inspired by non-responses related to political leaning being a problem in election surveys (Feldman, 2022; Clinton et al., 2022).

Baselines We compare the predictive performance of our fine-tuned LLM to several baselines. This allows us to explore to what extent and under which conditions fine-tuned LLMs might be more suitable for predicting survey responses than traditional classification approaches and zero-shot LLMs. We use the following baselines:

- Established tabular data classification methods: Logistic regression, Random forest, and CatBoost.
- Zero-shot predictions of GPT-4o-2024-11-20 as a state-of-the-art closed-source model baseline and our open-source model selection *without* fine-tuning.
- A majority class classifier. In our case, this means always predicting CDU/CSU – the party with the most votes in the German 2017 federal election.

Reproducibility All experiments are seeded. The dataset we use, GLES Cross-Section Cumulation 2009-2017, is openly available for researchers.⁷ Our code is available at <https://github.com/tobihol/llm-gles-finetuning>.

4 Results

4.1 EQ1: Missing Completely At Random

Figure 4 (a) and (b) show that fine-tuning outperforms zero-shot prediction for all model sizes and families. The state-of-the-art (SOTA) proprietary model, even though, performing better than most zero-shot models in (a) and all other zero-shot models in (b), is also outperformed significantly ($W > 1.98, p < .048$)⁸ for both EQ1 setups. The

difference between (a) and (b) is very noticeable, showing that zero-shot models struggle without a strong predictor as “party identification” (see Appendix 8). The fine-tuned open-source models perform similar to the classifier baselines.

The performance of the zero-shot models is not robust, as bigger models do not always outperform smaller models of the same family, e.g., for (a) Llama-3.1-8B performs significantly worse than Llama-3.2-3B ($W = -2.61, p = .009$) when used without fine-tuning. This is not the case for the fine-tuned models, where all models perform very similar across parameter range and model family.

Figure 5 (a) and (b) show that the zero-shot settings struggle more with modelling the vote share distribution of the GLES data. GPT4o has a tendency to predict more extreme parties to the right and to the left, and almost never predicting a person to be a Non-voter. Llama-3.1-8B zero-shot has a tendency to heavily over-predict left party share and, similarly almost no non-voter share. These tendencies can be explained through different factors, such as an over-prediction of parties that have been dominant in the training data (Santurkar et al., 2023) or a refusal to answer the prompt by the model (Wang et al., 2024). The fine-tuned models fit the vote share distribution well, similar to the baseline classifiers, both over-predicting the majority class slightly.

The fact that all classifier baselines and fine-tuned models, i.e., all models that have seen the training data, are at a similar performance level, suggests that for survey imputation where data is missing-completely-at-random, fine-tuning does not hold an advantage over traditional approaches. Note that this is the case only for scenarios similar to ours, where the data is well suited for traditional classification approaches.

4.2 EQ2a: Only Student Data

Not surprisingly the performance of all models drops when we use only survey responses of students to train and fine-tune the models (see Figure 4). However, the decrease in performance is far more pronounced for the baseline classifiers. The fine-tuned models retain more of their performance. Notably, the performance of the fine-tuned LLMs is not uniform as for EQ1. For this setting the smallest LLMs, i.e., Llama-3.2-1B and Qwen-2.5-0.5B, have variance in F1-scores showing instability of the fine-tuning process for these models. All fine-tuned models from the Llama family are able to

⁷<https://doi.org/10.4232/1.13648>

⁸Wilcoxon rank-sum tests indicated significant differences for all models variants at $\alpha = .05$.

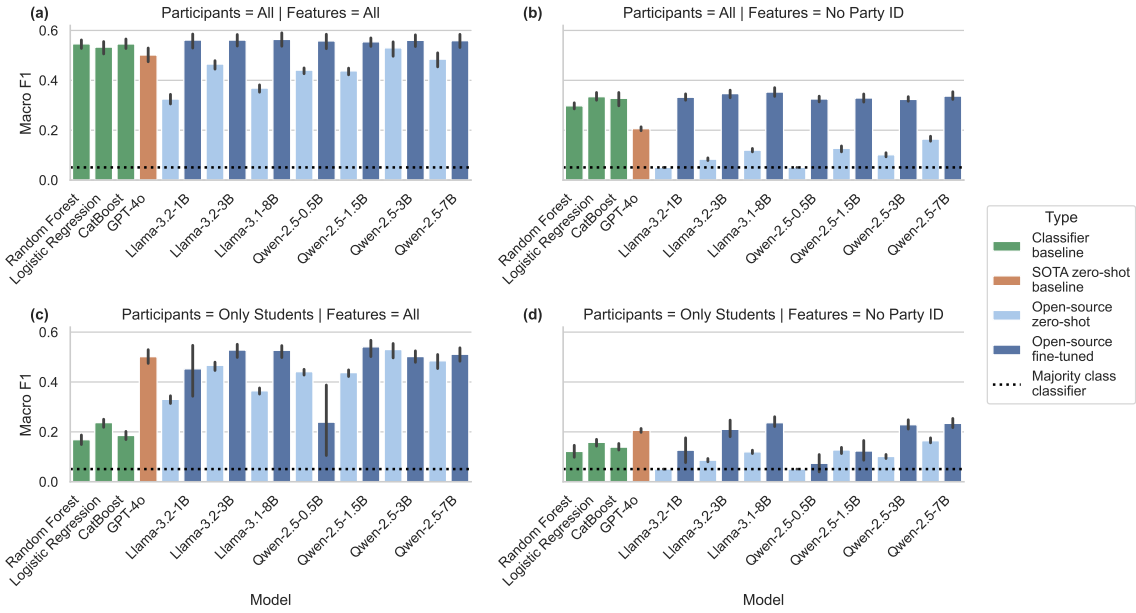


Figure 4: Performance of models when predicting survey responses. Subfigure (a) and (b) show the performance of models that are trained on responses of all participants that are not part of the randomly selected test set, while subfigure (c) and (d) show the performance of models that are trained on data from students only. Subplot (a) and (c) use all responses of participants to train models, while in subplot (b) and (d) the party identification is not included in the training data. One can see that fine-tuned LLMs outperform zero-shot LLMs and match the performance of traditional classification approaches when survey responses are missing at random. When systematic-non responses are present fine-tuned and zero-shot LLMs outperform traditional classification approaches. Fine-tuned LLMs seem to be slightly more robust to the missingness of features that are highly correlated with the variable that needs to be predicted.

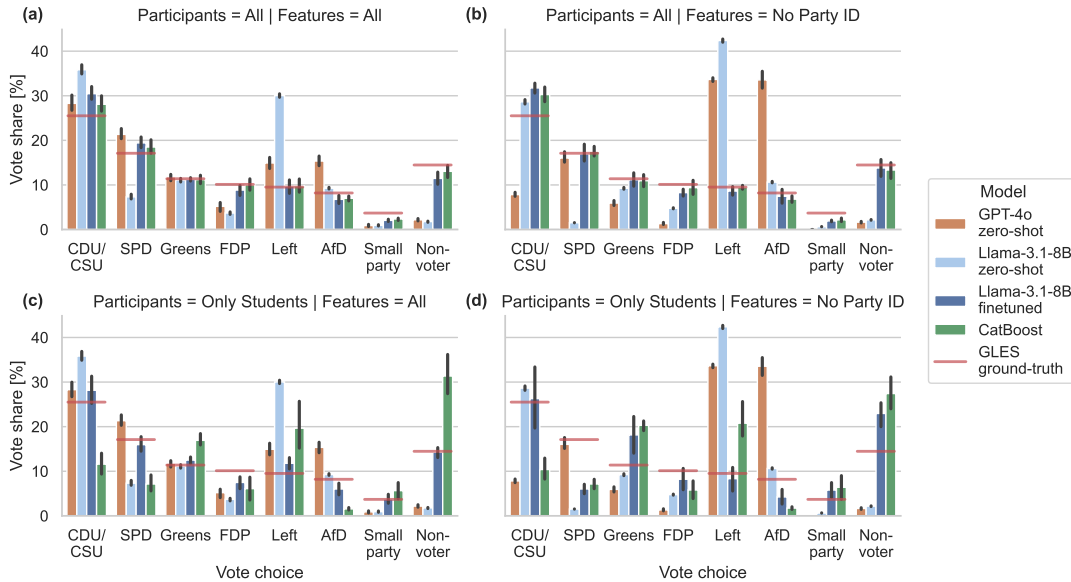


Figure 5: Performance of different models in predicting the vote share per party when vote-choices are missing at random (subplot (a) and (b)) and when only vote-choices of students were used to train models (subplot (c) and (d)). Subplot (a) and (b) show that especially the share of non-voters cannot be predicted with zero-shot LLMs. Also, for parties such as the FDP and AfD we see interesting differences between zero-shot LLMs and other methods. When “party identification” is removed the performance of zero-shot LLMs drops further. Subplot (c) and (d) show that fine-tuned LLMs and classification models trained on student’s responses only outperform zero-shot approaches, but tend to overestimate the share of left-leaning parties such as the left party and the greens. We also see that performance of fine-tuned LLMs is most robust to the removal of important features such as “party identification”. The distribution for other models and baselines show similar patterns, see Appendix B.3.

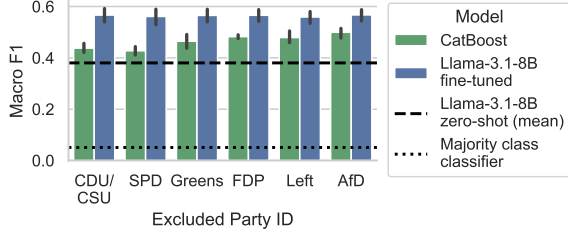


Figure 6: Performance of models that are trained on responses of all participants that are not part of the randomly selected test set and do not identify with one specific party that is depicted on the x-axis. Our results highlight that fine-tuned LLMs are the preferable method when systematic non-response bias is present. They outperform traditional classification approaches and zero-shot LLMs when people that identify with a selected party do not respond to the survey. We show the results for Llama-3.1-8B here, the other Llama models and baselines can be found in the Appendix Table 5 and show the same pattern.

outperform their zero-shot results, however this is not the case for the Qwen models, where the larger Qwen models slightly underperform their zero-shot scores for (c).

Even though GPT-4o shows similar macro F1 scores as our fine-tuned small models for EQ2a, the fine-tuned Llama-3.1-8B better represents the aggregated vote shares of the respondents. Especially for non-voters, extreme left (Left) and extreme right (AfD) parties, we see that GPT-4o produces larger deviations (see Figure 5).

4.3 EQ2b: Exclusion based on Party Identification

Figure 6 shows that the performance of fine-tuned LLMs is surprisingly robust to the exclusion of answers from survey participants that identify with one specific party. This is not the case for CatBoost, where the performance degrades relative to the size of participants that are excluded due to their party identification, e.g., the performance drop for excluding CDU/CSU is larger than the performance drop for excluding the AfD, because the vote share of CDU/CSU was bigger in 2017 (see Figure 5). Since fine-tuned LLMs are not only informed by the fine-tuning data, they also suffer less from biased data and are therefore more robust to non-response bias than traditional classification methods.

5 Discussion

The findings of this study contribute to the growing body of literature exploring the potential of LLMs in survey research, particularly in addressing missing data issues. Our study demonstrates that fine-tuned open-source LLMs are more effective in predicting voting behaviour than zero-shot approaches and can reduce political biases that may be present in LLMs. Further, our results indicate that supervised fine-tuning of LLMs can significantly enhance the robustness and performance of simulated survey responses, particularly in cases of systematic item non-response. Additionally, we observe that as LLMs increase in size, their fine-tuned performance tends to improve for systematic non-responses, suggesting the potential for further advancements in the future.

However, fine-tuning requires a minimum number of survey participants who provide complete responses to all relevant questions to achieve a significant performance gain over zero-shot methods. As a result, fine-tuned LLMs could facilitate the imputation of previously difficult-to-reconstruct survey data and pave the way for new planned missing-data survey designs, where only easy to reach sub-populations need to be sampled.

5.1 Limitations

Our work has several limitations that open up opportunities for future work. Firstly, we compare the performance of LLMs with traditional imputation methods. The most important imputation methods for dealing with item-non-response are multiple imputation and maximum likelihood estimations. In our work we focus on maximum likelihood estimations only since we are interested in producing imputed values that are as close as possible to the missing values. Secondly, we focused on single item-non-responses, while often multiple items (i.e. multiple survey answers of one participant) or units (i.e. all answers of one survey participant) are missing. Future work should investigate the ability of LLMs to simulate multiple responses of individuals.

Thirdly, our work uses zero-shot LLMs as baselines. However, we do not vary the prompts, and we acknowledge that the performance of zero-shot LLMs may change drastically when prompts are optimized.

Finally, this case study is limited to one important survey variable, vote choice. Future work

should explore if our results can be replicated when predicting other survey variables. While our work focused on fine-tuning LLMs, exploring other techniques like Retrieval-augmented Generation (RAG) or using the advanced in-context techniques enabled by the large context window of newer models would be an interesting option for future research.

5.2 Ethical Considerations

The data utilized in this study was derived from publicly available (upon request) survey results, with all responses anonymized to protect participants' privacy and confidentiality. The survey data does not contain any personally identifying information or offensive content.

An important ethical concern with simulating public opinions using Large Language Models is that the generated responses may inadvertently misrepresent genuine public sentiment. Because LLMs base their outputs on patterns found in historical and potentially biased data, there is a risk that they could reinforce existing prejudices, obscure minority perspectives, or amplify simplistic narratives, thereby offering a skewed view of societal attitudes. This study shows that fine-tuning might help to reduce these issues, but in no way mitigates them completely.

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A Additional Survey Data Information

A.1 Prompt Design

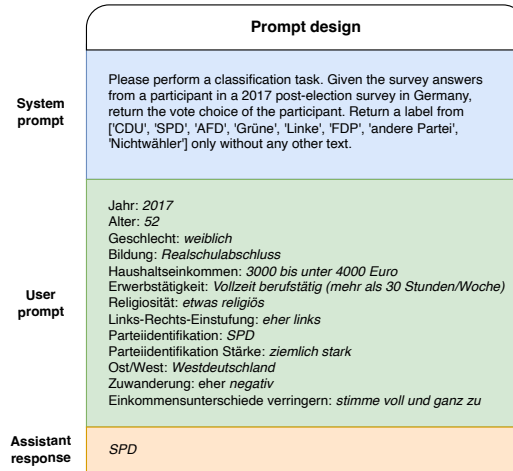


Figure 7: The original prompt design without translated survey items.

A.2 GLES Questionnaire

Variable	GLES Code	GLES Questionnaire (German)	GLES Questionnaire (English translation)
Year	year	Erhebungsjahr	Survey year*
Age	d2	Sagen Sie mir bitte, wie alt Sie sind.	Please tell me how old you are.*
Gender	d1	[Interviewer] Ist die Zielperson männlich oder weiblich?	[Interviewer] Is the respondent male or female?
Education	d5	Welchen höchsten allgemeinbildenden Schulabschluss haben Sie?	What's your highest level of general education?
Income	d43	Wie hoch ist das monatliche Netto-Einkommen IHRES HAUSHALTES INSGESAMT? Ich meine dabei die Summe, die nach Abzug von Steuern und Sozialversicherungsbeiträgen übrig bleibt. Bitte ordnen Sie Ihr Haushaltseinkommen in die Kategorien der Liste ein und nennen mir den Buchstaben.	Taken all together, would you please indicate what the monthly net income of your household is? By net income, I mean the amount that you have left after taxes and social security. Please select the monthly net income of your household from one of these groups and tell me the group letter.

Table 1: The GLES questions used for this study (Part 1). “GLES Code” and “GLES Questionnaire (German)” are taken directly from the GLES 2009-2017 Cross-section Cumulation codebook. An English translation of most of the questions was available in the GLES 2017 Cross-section Cumulation Documentation of Questionnaire. The translations with asterisks(*) were not available, but they were translated with DeepL.

Variable	GLES Code	GLES Questionnaire (German)	GLES Questionnaire (English translation)
Employment Status	d27	Nun weiter mit der Erwerbstätigkeit und Ihrem Beruf. Was von dieser Liste trifft auf Sie zu?	Do you currently work in a full-time or part-time job? Which of the descriptions in this list describes your status?
Religiosity	d26	Was würden Sie von sich sagen? Sind Sie überhaupt nicht religiös, nicht sehr religiös, etwas religiös oder sehr religiös?	What would you say about yourself, are you not religious at all, not very religious, somewhat religious or very religious?
Left-Right-Ideology	v74	Und wie ist das mit Ihnen selbst? Wo würden Sie sich auf der Skala von 1 bis 11 einordnen?	Where would you place yourself on this scale?
Party Identification	v25a	Und nun noch einmal kurz zu den politischen Parteien. In Deutschland neigen viele Leute längere Zeit einer bestimmten politischen Partei zu, obwohl sie auch ab und zu eine andere Partei wählen. Wie ist das bei Ihnen: Neigen Sie - ganz allgemein gesprochen - einer bestimmten Partei zu? Und wenn ja, welcher?	Now, let's look at the political parties. In Germany, many people lean towards a particular party for a long time, although they may occasionally vote for a different party. How about you, do you in general lean towards a particular party? If so, which one?
Party Identification Strength	v26	Wie stark oder wie schwach neigen Sie - alles zusammengefasst - dieser Partei zu: sehr stark, ziemlich stark, mäßig, ziemlich schwach oder sehr schwach?	All in all, how strongly or weakly do you lean toward this party: very strongly, fairly strongly, moderately, fairly weakly or very weakly?
Residency	ostwest2	Diese Variable gibt an, ob der/die Befragte in Ost- oder Westdeutschland lebt. Sie weist ost- und westdeutsche Befragte genau aus, d.h. innerhalb Berlins wurde nach (ehemaliger) Zugehörigkeit zur Deutschen Demokratischen Republik (DDR) und der Bundesrepublik Deutschland (BRD) unterschieden.	This variable indicates whether the respondent lives in East or West Germany. It precisely identifies East and West German respondents, i.e. within Berlin a distinction was made between (former) affiliation to the German Democratic Republic and the Federal Republic of Germany.*
Attitude towards Immigration	v88	Und wie ist Ihre Position zum Thema Zuzugsmöglichkeiten für Ausländer? Bitte benutzen Sie diese Skala.	And what position do you take on immigration for foreigners? Please use the scale.
Reducing inequality	v72d	Es gibt zu verschiedenen politischen Themen unterschiedliche Meinungen. Wie ist das bei Ihnen: Was halten Sie von folgenden Aussagen? Bitte antworten Sie anhand der Liste. (D) Die Regierung sollte Maßnahmen ergreifen, um die Einkommensunterschiede zu verringern.	There are various opinions on different political issues. What do you think of the following statements? Please use the list. (D) The government should take measures to reduce the differences in income levels.

Table 2: The GLES questions used for this study (Part 2).

A.3 Party Identification Impact

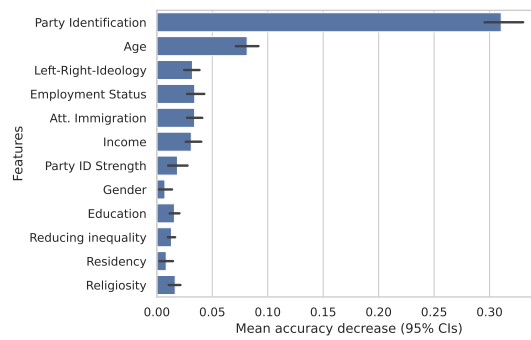


Figure 8: Feature importance for the Logistic Regression classification in EQ1 computed using feature permutation, where each feature's values are randomly shuffled to quantify the drop in performance. This drop indicates how much each feature contributes to the model's predictive power.

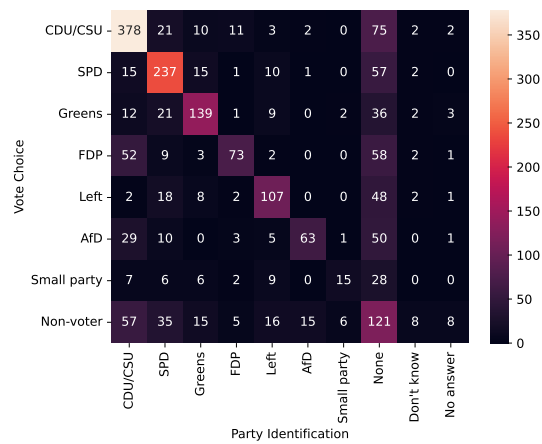


Figure 9: Cross-tabulation of “vote choice” and “party identification”.

B Results Appendix

B.1 Other Performance Metrics

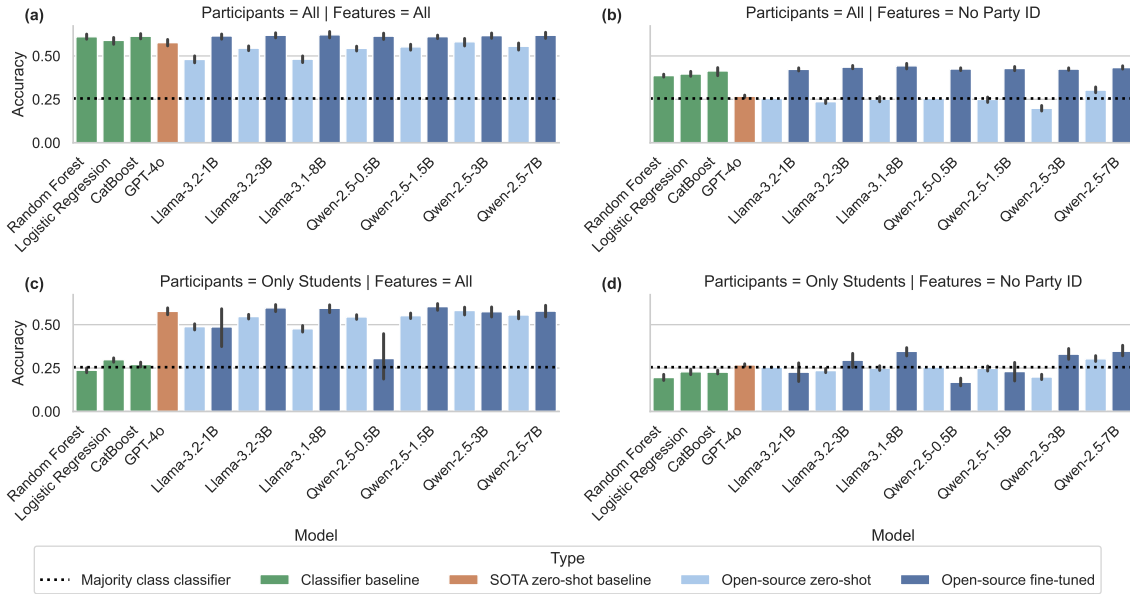


Figure 10: Accuracy scores of models when predicting survey responses for EQ1 and EQ2a.

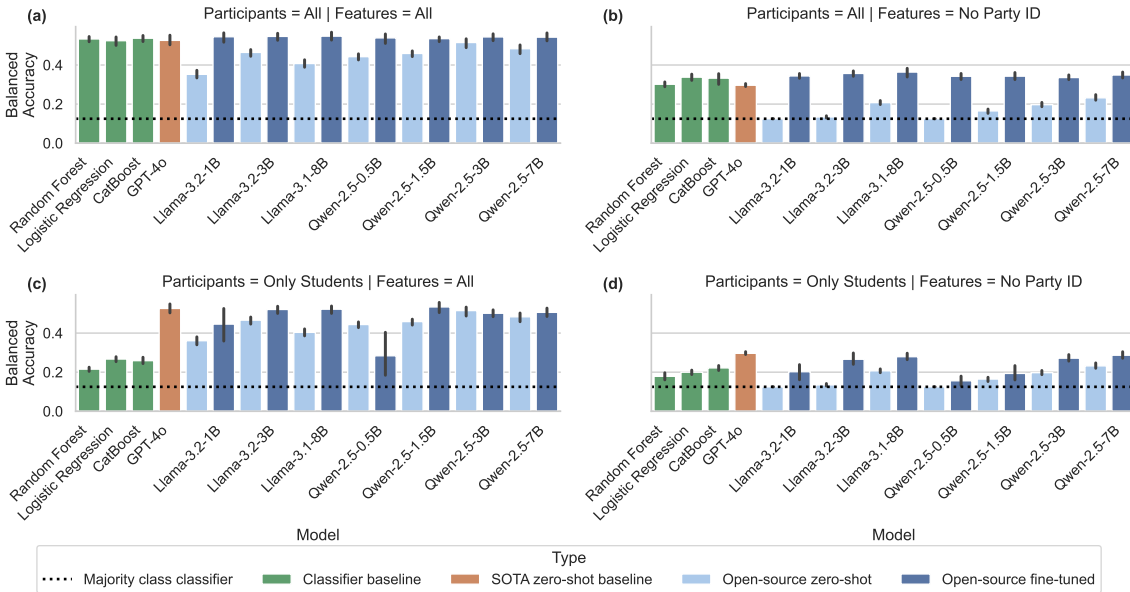


Figure 11: Balanced accuracy scores of models when predicting survey responses for EQ1 and EQ2a.

B.2 Results as Tabular Data

Participants Features Model	All All	All No Party ID	Only Students All	Only Students No Party ID
Random Forest	0.546 [0.528, 0.562]	0.298 [0.285, 0.310]	0.169 [0.149, 0.187]	0.122 [0.098, 0.146]
Logistic Regression	0.533 [0.506, 0.555]	0.334 [0.320, 0.351]	0.237 [0.218, 0.251]	0.158 [0.143, 0.170]
CatBoost	0.546 [0.528, 0.566]	0.328 [0.298, 0.351]	0.186 [0.169, 0.202]	0.139 [0.127, 0.153]
GPT-4o zero-shot	0.501 [0.475, 0.530]	0.206 [0.198, 0.214]	0.501 [0.475, 0.530]	0.206 [0.198, 0.214]
Llama-3.2-1B zero-shot	0.325 [0.305, 0.344]	0.051 [0.051, 0.051]	0.330 [0.314, 0.345]	0.051 [0.051, 0.051]
Llama-3.2-1B fine-tuned	0.561 [0.529, 0.586]	0.332 [0.320, 0.346]	0.453 [0.343, 0.547]	0.126 [0.077, 0.176]
Llama-3.2-3B zero-shot	0.465 [0.445, 0.479]	0.084 [0.078, 0.089]	0.467 [0.446, 0.480]	0.086 [0.080, 0.093]
Llama-3.2-3B fine-tuned	0.561 [0.537, 0.584]	0.347 [0.330, 0.361]	0.528 [0.498, 0.551]	0.210 [0.181, 0.247]
Llama-3.1-8B zero-shot	0.369 [0.352, 0.382]	0.120 [0.113, 0.127]	0.365 [0.351, 0.376]	0.120 [0.113, 0.127]
Llama-3.1-8B fine-tuned	0.564 [0.537, 0.590]	0.353 [0.335, 0.371]	0.527 [0.498, 0.546]	0.237 [0.221, 0.261]
Qwen-2.5-0.5B zero-shot	0.441 [0.426, 0.450]	0.051 [0.051, 0.051]	0.441 [0.428, 0.451]	0.051 [0.051, 0.051]
Qwen-2.5-1.5B zero-shot	0.438 [0.422, 0.449]	0.127 [0.113, 0.137]	0.438 [0.422, 0.449]	0.127 [0.113, 0.138]
Qwen-2.5-3B zero-shot	0.530 [0.496, 0.554]	0.101 [0.093, 0.110]	0.530 [0.496, 0.554]	0.101 [0.093, 0.109]
Qwen-2.5-7B zero-shot	0.485 [0.454, 0.510]	0.164 [0.156, 0.176]	0.485 [0.454, 0.510]	0.164 [0.156, 0.176]
Qwen-2.5-0.5B fine-tuned	0.558 [0.527, 0.585]	0.325 [0.314, 0.337]	0.239 [0.105, 0.388]	0.074 [0.039, 0.109]
Qwen-2.5-1.5B fine-tuned	0.554 [0.536, 0.570]	0.329 [0.313, 0.345]	0.541 [0.502, 0.567]	0.123 [0.086, 0.165]
Qwen-2.5-3B fine-tuned	0.560 [0.535, 0.582]	0.323 [0.316, 0.334]	0.502 [0.479, 0.525]	0.229 [0.211, 0.248]
Qwen-2.5-7B fine-tuned	0.559 [0.532, 0.584]	0.337 [0.323, 0.354]	0.511 [0.483, 0.537]	0.234 [0.217, 0.254]

Table 3: Macro F1 scores of models when predicting survey responses for EQ1 and EQ2a.

Participants	Features	Model Vote Choice	GPT-4o zero-shot	Llama-3.1-8B zero-shot	Llama-3.1-8B fine-tuned	CatBoost
All	All	AfD	0.154 [0.142, 0.165]	0.093 [0.091, 0.095]	0.068 [0.059, 0.079]	0.070 [0.064, 0.076]
		CDU/CSU	0.283 [0.267, 0.302]	0.358 [0.348, 0.369]	0.305 [0.291, 0.319]	0.281 [0.264, 0.298]
		FDP	0.052 [0.041, 0.060]	0.037 [0.034, 0.039]	0.088 [0.075, 0.099]	0.100 [0.088, 0.114]
		Greens	0.118 [0.109, 0.124]	0.111 [0.107, 0.115]	0.112 [0.109, 0.116]	0.112 [0.103, 0.122]
		Left	0.149 [0.137, 0.163]	0.300 [0.297, 0.305]	0.097 [0.082, 0.114]	0.098 [0.084, 0.113]
		Non-voter	0.022 [0.019, 0.025]	0.018 [0.016, 0.019]	0.115 [0.100, 0.129]	0.131 [0.116, 0.144]
		SPD	0.213 [0.203, 0.226]	0.073 [0.069, 0.079]	0.194 [0.183, 0.209]	0.185 [0.171, 0.201]
		Small party	0.009 [0.006, 0.012]	0.009 [0.008, 0.011]	0.021 [0.018, 0.023]	0.024 [0.020, 0.026]
		No Party ID	0.336 [0.316, 0.355]	0.106 [0.105, 0.108]	0.075 [0.059, 0.090]	0.068 [0.061, 0.076]
		CDU/CSU	0.078 [0.074, 0.083]	0.286 [0.281, 0.292]	0.318 [0.304, 0.328]	0.303 [0.286, 0.318]
Only Students	All	FDP	0.013 [0.011, 0.016]	0.048 [0.046, 0.049]	0.083 [0.075, 0.090]	0.094 [0.079, 0.111]
		Greens	0.059 [0.054, 0.065]	0.093 [0.090, 0.094]	0.112 [0.096, 0.127]	0.109 [0.096, 0.125]
		Left	0.337 [0.332, 0.340]	0.424 [0.420, 0.427]	0.086 [0.077, 0.096]	0.095 [0.090, 0.100]
		Non-voter	0.017 [0.015, 0.018]	0.022 [0.021, 0.022]	0.138 [0.119, 0.157]	0.133 [0.114, 0.150]
		SPD	0.160 [0.151, 0.175]	0.015 [0.015, 0.015]	0.170 [0.152, 0.189]	0.175 [0.165, 0.187]
		Small party	0.000	0.006 [0.006, 0.007]	0.019 [0.017, 0.021]	0.022 [0.018, 0.026]
		AfD	0.154 [0.142, 0.165]	0.093 [0.091, 0.095]	0.060 [0.045, 0.074]	0.016 [0.013, 0.019]
		CDU/CSU	0.283 [0.267, 0.302]	0.358 [0.348, 0.369]	0.282 [0.252, 0.311]	0.116 [0.092, 0.140]
		FDP	0.052 [0.041, 0.060]	0.037 [0.034, 0.039]	0.075 [0.062, 0.089]	0.061 [0.039, 0.090]
		Greens	0.118 [0.109, 0.124]	0.111 [0.107, 0.115]	0.125 [0.117, 0.132]	0.169 [0.159, 0.184]
Only Students	All	Left	0.149 [0.137, 0.163]	0.300 [0.297, 0.305]	0.118 [0.105, 0.131]	0.197 [0.152, 0.257]
		Non-voter	0.022 [0.019, 0.025]	0.018 [0.016, 0.019]	0.143 [0.132, 0.153]	0.314 [0.273, 0.364]
		SPD	0.213 [0.203, 0.226]	0.073 [0.069, 0.079]	0.160 [0.145, 0.178]	0.071 [0.056, 0.092]
		Small party	0.009 [0.006, 0.012]	0.009 [0.008, 0.011]	0.038 [0.028, 0.048]	0.057 [0.037, 0.076]
		AfD	0.336 [0.316, 0.355]	0.106 [0.105, 0.108]	0.042 [0.026, 0.059]	0.017 [0.014, 0.021]
		CDU/CSU	0.078 [0.074, 0.083]	0.286 [0.281, 0.292]	0.263 [0.192, 0.329]	0.104 [0.081, 0.129]
		FDP	0.013 [0.011, 0.016]	0.048 [0.046, 0.049]	0.082 [0.058, 0.106]	0.058 [0.039, 0.080]
		Greens	0.059 [0.054, 0.065]	0.093 [0.090, 0.094]	0.182 [0.141, 0.227]	0.203 [0.190, 0.213]
		Left	0.337 [0.332, 0.340]	0.424 [0.420, 0.427]	0.083 [0.054, 0.110]	0.208 [0.179, 0.256]
		Non-voter	0.017 [0.015, 0.018]	0.022 [0.021, 0.022]	0.230 [0.202, 0.254]	0.274 [0.234, 0.318]
Only Students	All	SPD	0.160 [0.151, 0.175]	0.015 [0.015, 0.015]	0.060 [0.050, 0.071]	0.071 [0.061, 0.082]
		Small party	0.000	0.006 [0.006, 0.007]	0.058 [0.041, 0.075]	0.064 [0.037, 0.089]

Table 4: Aggregated vote shares [0, 1] of models when predicting survey responses for EQ1 and EQ2a.

Excluded Party ID Model	CDU/CSU	SPD	Greens	FDP	Left	AfD
Random Forest	0.422 [0.399, 0.439]	0.428 [0.403, 0.443]	0.443 [0.423, 0.463]	0.487 [0.463, 0.509]	0.466 [0.448, 0.482]	0.494 [0.474, 0.510]
Logistic Regression	0.445 [0.414, 0.469]	0.443 [0.419, 0.463]	0.470 [0.439, 0.501]	0.486 [0.467, 0.502]	0.476 [0.452, 0.500]	0.510 [0.489, 0.527]
CatBoost	0.438 [0.421, 0.457]	0.427 [0.411, 0.444]	0.465 [0.437, 0.491]	0.483 [0.475, 0.490]	0.479 [0.459, 0.505]	0.499 [0.477, 0.515]
Llama-3.2-1B zero-shot	0.313 [0.293, 0.331]	0.310 [0.290, 0.327]	0.310 [0.292, 0.327]	0.310 [0.292, 0.327]	0.310 [0.292, 0.327]	0.312 [0.296, 0.326]
Llama-3.2-1B fine-tuned	0.548 [0.523, 0.571]	0.558 [0.531, 0.581]	0.560 [0.534, 0.579]	0.563 [0.532, 0.588]	0.558 [0.526, 0.585]	0.560 [0.531, 0.583]
Llama-3.2-3B zero-shot	0.468 [0.447, 0.480]	0.467 [0.447, 0.480]	0.468 [0.447, 0.480]	0.468 [0.447, 0.480]	0.468 [0.447, 0.480]	0.466 [0.445, 0.480]
Llama-3.2-3B fine-tuned	0.558 [0.529, 0.583]	0.558 [0.533, 0.580]	0.569 [0.541, 0.594]	0.557 [0.530, 0.580]	0.572 [0.546, 0.595]	0.561 [0.538, 0.583]
Llama-3.1-8B zero-shot	0.369 [0.354, 0.382]	0.364 [0.351, 0.376]	0.369 [0.352, 0.382]	0.368 [0.352, 0.380]	0.369 [0.352, 0.382]	0.369 [0.352, 0.382]
Llama-3.1-8B fine-tuned	0.567 [0.540, 0.592]	0.561 [0.529, 0.590]	0.565 [0.540, 0.589]	0.566 [0.542, 0.588]	0.559 [0.534, 0.580]	0.567 [0.543, 0.588]
Qwen-2.5-0.5B zero-shot	0.441 [0.428, 0.449]	0.441 [0.426, 0.450]	0.441 [0.426, 0.450]	0.441 [0.426, 0.450]	0.441 [0.426, 0.450]	0.441 [0.426, 0.450]
Qwen-2.5-0.5B fine-tuned	0.552 [0.527, 0.577]	0.557 [0.534, 0.578]	0.558 [0.525, 0.586]	0.560 [0.530, 0.589]	0.556 [0.528, 0.579]	0.557 [0.532, 0.582]
Qwen-2.5-1.5B zero-shot	0.438 [0.423, 0.449]	0.438 [0.422, 0.449]	0.438 [0.422, 0.449]	0.438 [0.422, 0.449]	0.438 [0.422, 0.449]	0.438 [0.422, 0.449]
Qwen-2.5-1.5B fine-tuned	0.553 [0.526, 0.578]	0.555 [0.527, 0.580]	0.555 [0.524, 0.583]	0.558 [0.522, 0.588]	0.559 [0.524, 0.591]	0.551 [0.521, 0.577]
Qwen-2.5-3B zero-shot	0.530 [0.496, 0.554]	0.530 [0.496, 0.554]	0.530 [0.496, 0.554]	0.530 [0.496, 0.554]	0.530 [0.496, 0.554]	0.530 [0.496, 0.554]
Qwen-2.5-3B fine-tuned	0.559 [0.533, 0.583]	0.565 [0.536, 0.592]	0.558 [0.534, 0.579]	0.566 [0.535, 0.595]	0.558 [0.532, 0.580]	0.562 [0.536, 0.585]
Qwen-2.5-7B zero-shot	0.485 [0.454, 0.510]	0.485 [0.454, 0.510]	0.485 [0.454, 0.510]	0.485 [0.454, 0.510]	0.485 [0.454, 0.510]	0.485 [0.454, 0.510]
Qwen-2.5-7B fine-tuned	0.557 [0.530, 0.582]	0.565 [0.541, 0.590]	0.561 [0.539, 0.583]	0.560 [0.528, 0.590]	0.566 [0.538, 0.592]	0.568 [0.537, 0.595]

Table 5: Macro F1 scores of models in EQ2b.

B.3 Vote share distribution for other models

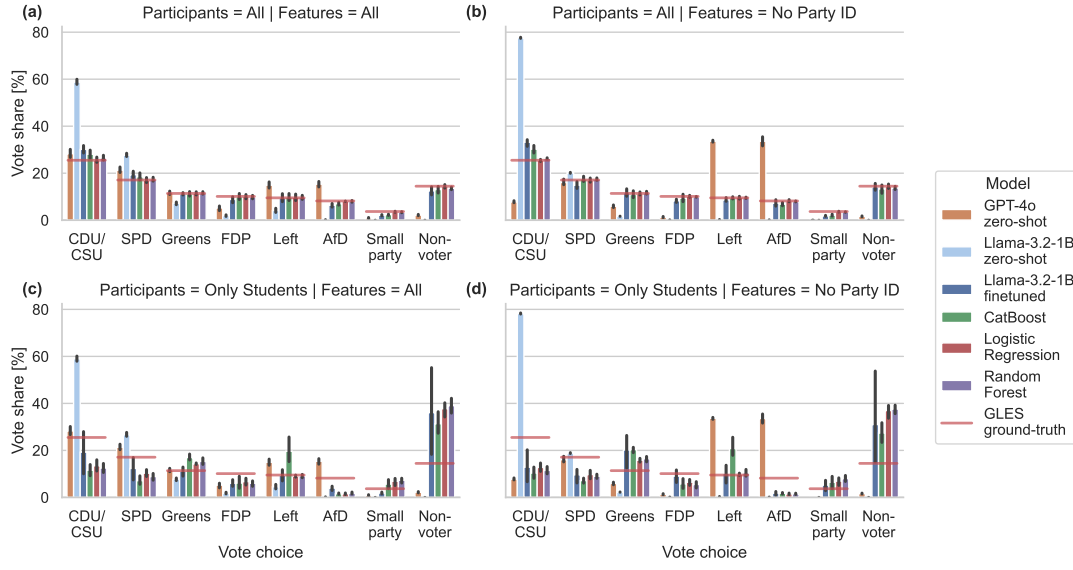


Figure 12: Vote share distribution for Llama-3.2-1B in EQ1 and EQ2a.

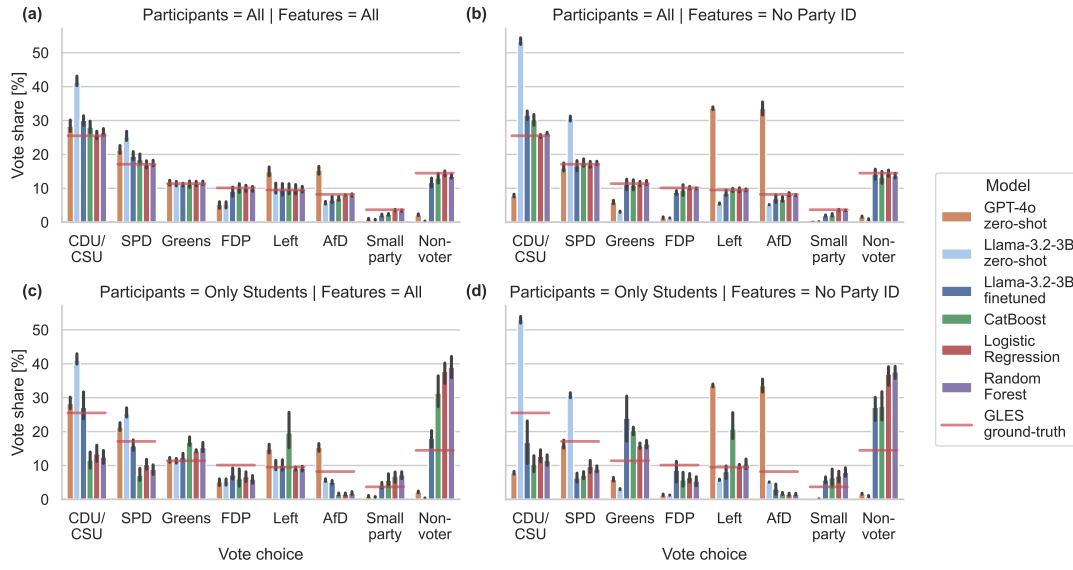


Figure 13: Vote share distribution for Llama-3.2-3B in EQ1 and EQ2a.

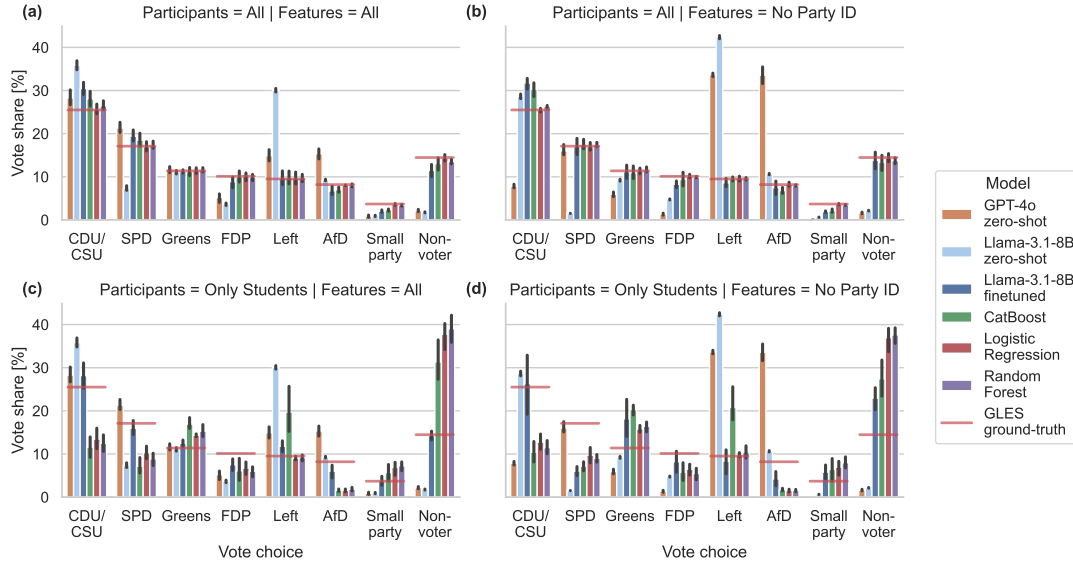


Figure 14: Vote share distribution for Llama-3.1-8B in EQ1 and EQ2a.

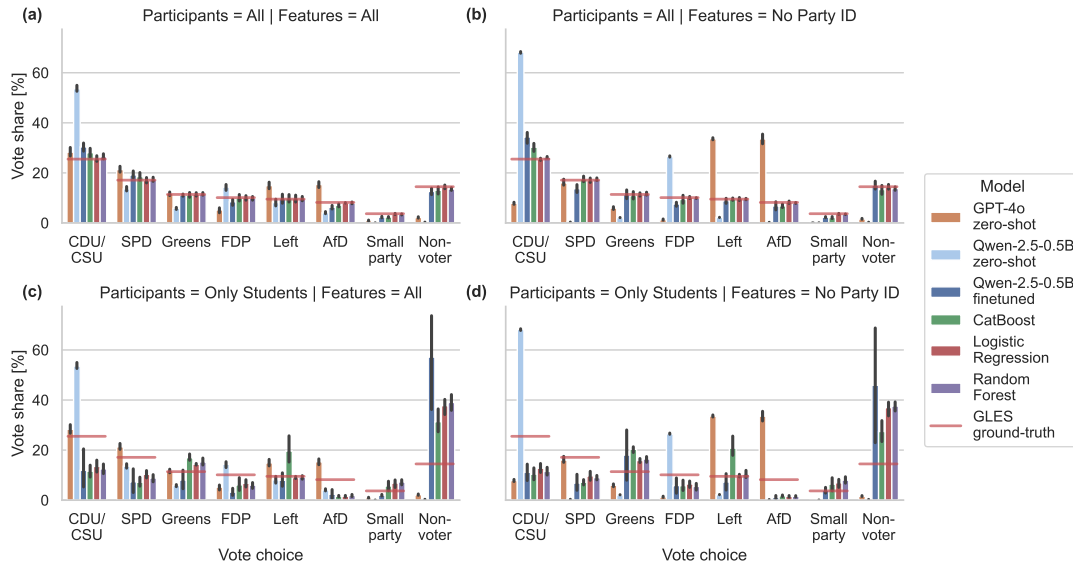


Figure 15: Vote share distribution for Qwen-2.5-0.5B in EQ1 and EQ2a.

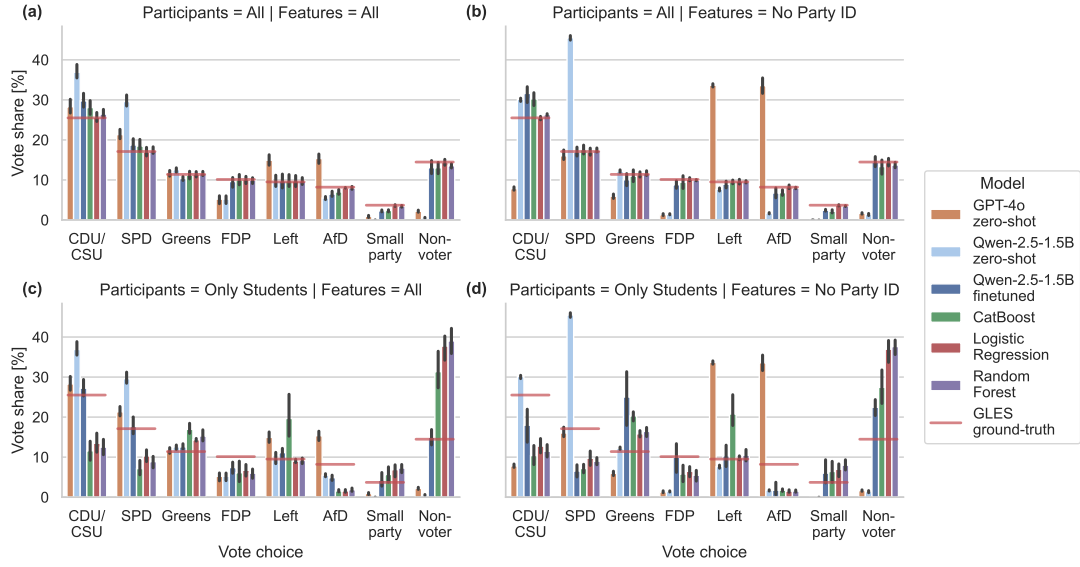


Figure 16: Vote share distribution for Qwen-2.5-1.5B in EQ1 and EQ2a.

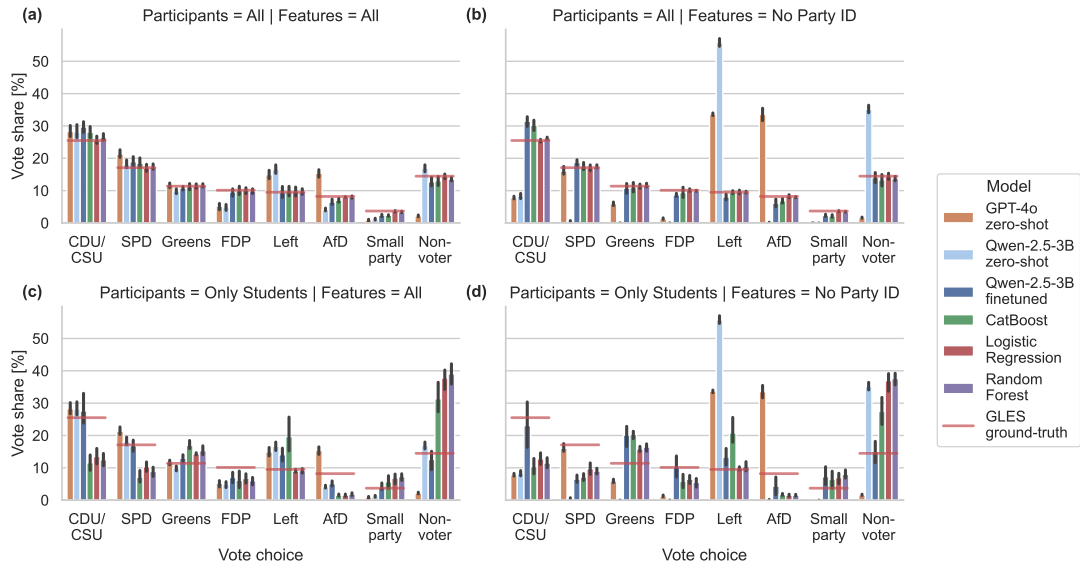


Figure 17: Vote share distribution for Qwen-2.5-3B in EQ1 and EQ2a.

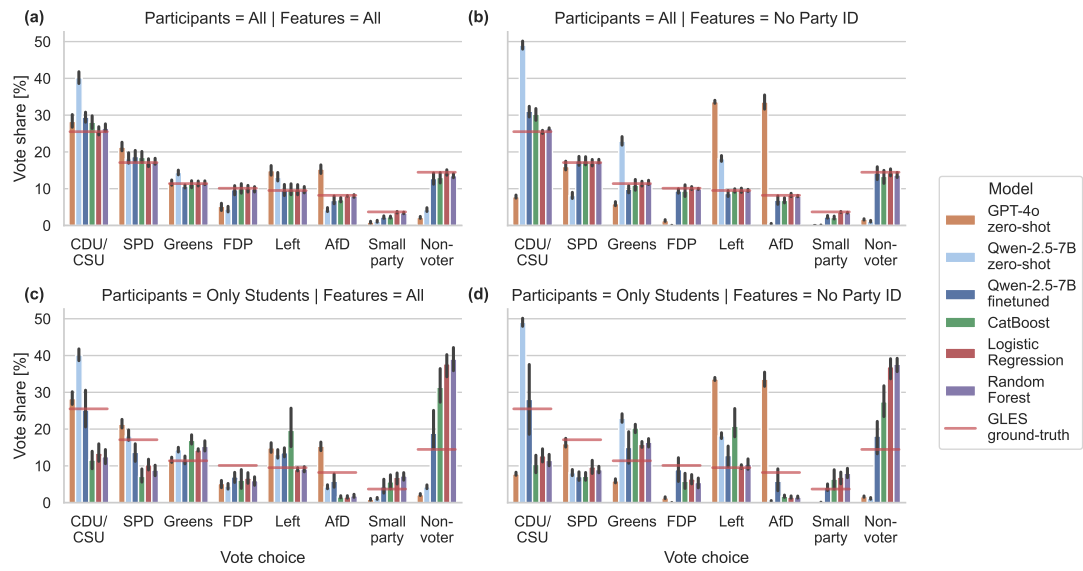


Figure 18: Vote share distribution for Qwen-2.5-7B in EQ1 and EQ2a.